

Quantum Hybrid Image Transfer via Superdense Coding

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Abstract—In this paper, we present a new hybrid image transmission method that combines classical and quantum for effective and reliable image delivery. We start by dividing images into blocks and calculating the entropy of each block to find areas with high information content. We transmit high-entropy blocks, using SuperDense Coding. The remaining blocks and important metadata are sent through classical channels. The receiver uses the metadata to rebuild the original image and applies error recovery methods to cope with channel noise and quantum transmission errors. Experimental results show Structural Similarity Index Measure(SSIM) of 0.996 and Image Fidelity of 99.96%. In our prototype we focus on the quantum SDC-based transmission, since classical transmission is already well-established and operational. Some limitations and future research areas include managing quantum entanglement, reducing errors, and scaling SDC for larger uses.

I Introduction

Image transmission plays a crucial role in modern communication systems. Its applications range from sharing on social media to satellite imaging, surveillance, and telemedicine. As high-resolution imaging grows, so does the need for real-time communication. Transmitting large data efficiently and securely has become a significant challenge. Currently, traditional communication systems dominate this area. They convert images into sequences of bits (0s and 1s), which are then compressed using techniques like JPEG, Huffman coding, or run-length encoding. To ensure reliable data transfer, methods like Forward Error Correction (FEC) or retransmission protocols are used. The data travels through common channels such as wired internet, Wi-Fi, or wireless cellular networks. While traditional transmission is robust and scalable, it is limited by bandwidth availability and can be vulnerable to data interception or modification.

In contrast, quantum communication offers new ways to transfer data. A notable example is Superdense Coding(SDC). This protocol allows the transmission of two classical bits of information using just one qubit, assuming that Alice and Bob share entanglement beforehand. While traditional

communication needs two separate bits sent over a channel, SDC encodes them into a single qubit transmission. This effectively doubles the channel capacity and offers inherent quantum security. [1]

Although SDC is theoretically powerful, transmitting an entire image through SDC alone is not practical today. Current quantum systems have issues like limited qubit counts, noisy quantum gates, decoherence, and challenges in maintaining entanglement over long distances. These limitations block the large-scale use of SDC for multimedia files. [2], [3]

To fill this gap, we suggest a hybrid image transmission approach. In this method, the image is divided into blocks and scored by entropy. Low-entropy blocks (backgrounds or repetitive areas) are sent via the classical channel, while high-entropy blocks (edges and details) are transmitted through the quantum channel using SDC. This ensures that quantum resources are utilized only where they are most beneficial.

This hybrid strategy merges the scalability and reliability of traditional transmission with the bandwidth efficiency and security of quantum SDC. It represents a practical advancement given the current quantum hardware limitations. This approach lays the groundwork for future communication systems that can integrate classical and quantum technologies.

II Literature Review

Classical communication achieves efficiency through compression and channel coding techniques. Compression methods like Huffman and arithmetic remove redundancy in image data while Forward error correction helps deal with noise, making long-distance connections reliable. Despite these advances, classical communication remains fundamentally constrained by Shannon's channel capacity, which limits the maximum achievable information rate based on channel bandwidth and noise characteristics. [4], [5]

This is where superdense coding (SDC) shows its benefits over purely classical methods by enabling the transmission of two classical bits using a single qubit, assuming prior

entanglement between sender and receiver [1]. SDC effectively increases the classical payload per channel use without requiring additional bandwidth [6], [7]. This makes SDC a complementary enhancement to classical methods, boosting throughput while remaining compatible with quantum-secure transmission.

However, existing research on SDC and related quantum communication techniques focuses mainly on theoretical capacity analysis or small-scale demonstrations. The direct application of SDC to large multimedia data, such as full-resolution images, is limited by current quantum hardware constraints, including restricted qubit counts, gate noise, and decoherence. Consequently, purely quantum approaches remain impractical for real-world image transmission scenarios.

This perspective forms the basis for the hybrid image transmission framework proposed in this work. By combining SDC with established classical compression and error correction techniques, hybrid approaches aim to leverage the strengths of both paradigms.

III Methodology and Approach

When multiple communication paths operate in parallel, the overall effective throughput is the sum of the individual channel throughputs [5]. In 5G networks, devices often use multi-connectivity, simultaneously combining WiFi, LTE, and 5G carriers, demonstrating that the effective data rate in such systems can be calculated as the sum of the individual link rates, reflecting independent and concurrent contributions from each channel [8]. A similar approach is used in hybrid Free-Space Optical (FSO) and Radio Frequency (RF) systems, where traffic may be switched or multiplexed across optical and radio links [9]. Integrated satellite-5G architectures follow the same principle, where satellite and terrestrial paths jointly contribute to system capacity [10].

These established models provide the theoretical motivation for treating classical and quantum (SDC-based) transmission paths in our hybrid framework as parallel information channels, whose effective throughput is naturally represented as the sum of their respective contributions.

Throughput Advantage of Hybrid Approach

According to Shannon's theorem [4], the capacity of a classical communication channel of bandwidth B and signal-to-noise ratio (S/N) is given by:

$$C_{\text{classical}} = B \log_2 \left(1 + \frac{S}{N} \right) \quad (1)$$

Using Superdense Coding (SDC) [1], it is possible to transmit two classical bits per qubit. Hence, the effective SDC channel capacity becomes:

$$C_{\text{SDC}} = 2 \times C_{\text{classical}} \quad (2)$$

In the proposed hybrid transmission model, only an α -fraction ($0 \leq \alpha \leq 1$) of the image blocks are transmitted via the quantum (SDC) channel, while the remaining $(1 - \alpha)$

fraction uses the classical channel. Therefore, the total hybrid channel capacity can be expressed as a weighted sum:

$$C_{\text{hybrid}} = (1 - \alpha)C_{\text{classical}} + \alpha C_{\text{SDC}} \quad (3)$$

Substituting Eq. (2) into Eq. (3):

$$\begin{aligned} C_{\text{hybrid}} &= (1 - \alpha)C_{\text{classical}} + \alpha(2C_{\text{classical}}) \\ &= (1 - \alpha + 2\alpha)C_{\text{classical}} \\ &= (1 + \alpha)C_{\text{classical}} \end{aligned} \quad (4)$$

Finally, substituting Eq. (1) into Eq. (4) gives the closed-form expression for hybrid channel capacity:

$$C_{\text{hybrid}} = (1 + \alpha)B \log_2 \left(1 + \frac{S}{N} \right) \quad (5)$$

Example

To show the improvement achieved by our hybrid model, consider a classical channel of bandwidth $B = 1$ MHz and signal-to-noise ratio $S/N = 15$ dB, corresponding to $S/N = 31.6$ in linear scale.

From Eq. (1), the Shannon capacity is:

$$\begin{aligned} C_{\text{classical}} &= B \log_2 \left(1 + \frac{S}{N} \right) \\ &= 10^6 \log_2(1 + 31.6) \\ &= 5.02 \approx 5 \text{ Mbps} \end{aligned} \quad (6)$$

For a quantum allocation fraction of $\alpha = 0.2$ (i.e., 20% of high-entropy blocks transmitted via SDC), the hybrid capacity from Eq. (5) becomes:

$$\begin{aligned} C_{\text{hybrid}} &= (1 + \alpha)B \log_2 \left(1 + \frac{S}{N} \right) \\ &= 1.15 \times 10^6 \log_2(1 + 31.6) \\ &= 5.78 \approx 5.8 \text{ Mbps} \end{aligned} \quad (7)$$

$$\frac{C_{\text{hybrid}} - C_{\text{classical}}}{C_{\text{classical}}} \times 100\% = 16\%$$

This capacity gain arises from entanglement assistance and does not contradict Shannon's classical capacity theorem, which assumes unentangled classical channels.

III-A Conceptual Framework

Our system is a hybrid transmission model that uses both classical and quantum channels. The classical channel acts as a reliable, high-throughput medium for bulk image transfer, especially for low-information or repetitive blocks. It supports well-known techniques like compression, error correction, and retransmission.

By using superdense coding (SDC), each qubit encodes two classical bits, which doubles channel efficiency compared to classical transfer. This division of tasks between channels is key to our approach, ensuring reliability and efficiency.

III-B Workflow Overview

The overall workflow starts with image preprocessing. The image is divided into blocks of non-overlapping size $B \times B$. Smaller blocks improve localization, while larger blocks reduce metadata overhead. For an image sized $H \times W$, the total number of blocks is

$$M = \text{rows} \times \text{cols}.$$

Each block is scored using Shannon entropy to estimate its information content. We select the blocks with the highest entropy for SDC transmission, while the others are set for classical transfer. Shannon entropy [4], [5] :

$$H_{\text{block}} = - \sum_i p(i) \log_2 p(i).$$

where $p(i)$ is the probability distribution of pixel intensities in the block.

Based on entropy scores, we selected the top- k of blocks with the highest information content for quantum transmission using SDC. The indices of these selected blocks were recorded in the metadata.

III-C Design Choices and Policies

A key design choice in our methodology is entropy-based block selection. This approach ensures that limited quantum resources go to the most informative areas of the image, improving transmission efficiency. In our prototype, we used a top- $p\%$ selection policy, where the highest-entropy fraction of the blocks goes to the quantum channel.

In our prototype, the metadata was kept minimal but enough to reconstruct the image layout. The information is stored in a structured format with two main components:

- Global metadata fields:
 - Block size used for partitioning (e.g., 16×16)
 - Top percentage parameter indicating the fraction of blocks selected for quantum transmission
- Per-block entries:
 - Row and column indices of each block
 - Block type: quantum (selected high-entropy blocks) or classical (remaining blocks)

Existing quantum image representations such as FRQI [11] and NEQR [12] lack scalable classical-quantum interfaces and suffer from fundamental state-preparation and measurement limitations, which restrict their applicability largely to simulation-based studies under current hardware constraints [13].

Motivated by these limitations, a key design choice in our methodology is to retain image data in the classical domain and map classical bitstreams to qubits exclusively for transmission using Superdense Coding, thereby exploiting quantum communication advantages without incurring the overhead of full quantum image encoding.

IV Prototype Development

IV-A Prototype Description

The system is entirely implemented in Python, combining classical image processing libraries with Qiskit for quantum simulation.

To reduce simulation runtime, we executed block-wise SDC encoding in parallel with Python multiprocessing as each block is independent of other. This allowed multiple blocks to be transmitted at the same time in Qiskit. Although this optimization speeds up the prototype, it is not necessary in real quantum hardware, where qubits are processed at the same time on the physical level.

IV-B Components and Tools

- **Qiskit:** Used to build and simulate SDC circuits, prepare entanglement, encode using Pauli gates, and perform Bell measurements.
- **NumPy/OpenCV:** Handles image preprocessing, calculating entropy, dividing blocks, compressing, and evaluating PSNR/SSIM.
- **Python:** Serves as the primary programming environment.

IV-C Prototype Workflow:

- 1) **Input loading:** The system loads the input image for processing.
- 2) **Block partitioning and entropy computation:** The image is divided into non-overlapping blocks, and entropy is calculated for each block to measure information content.
- 3) **Block selection:** High-entropy blocks are assigned to the quantum (SDC) channel, while low-entropy blocks are designated for classical transfer.
- 4) **Encoding and transmission:** High-entropy blocks are encoded using Qiskit-based SDC circuits, while classical blocks are handled through conventional array-based transmission.
- 5) **Decoding:** At the receiver, both streams are decoded using metadata to ensure correct block placement.
- 6) **Reconstruction and evaluation:** The image is reconstructed from all blocks and evaluated using quality metrics such as PSNR and SSIM.

IV-D Performance Evaluation

The computational cost of the proposed hybrid image transmission framework primarily depends on three stages: *block-wise entropy computation*, *block selection*, and *superdense coding (SDC) simulation*.

a) Entropy Computation:

Each image of size $H \times W$ is divided into blocks of $B \times B$. The number of blocks is $M = \frac{H \times W}{B^2}$. Computing Shannon entropy for each block requires scanning all B^2 pixels, giving a time complexity of:

$$O(M \times B^2) = O(HW)$$

which scales linearly with the number of pixels.

b) *Block Selection:*

Sorting the entropy scores to select the top- $p\%$ blocks has a complexity of:

$$O(M \log M)$$

though this cost is small relative to the image size for practical block counts.

c) *SDC Simulation:*

Each 2-bit symbol is encoded via an SDC circuit. Assuming Q qubits (or bit pairs) are processed, the quantum encoding and simulation in Qiskit has an empirical complexity of approximately:

$$O(Q)$$

per circuit generation, but with a large constant overhead due to quantum gate simulation. Parallel execution reduces effective runtime by a factor of k , the number of CPU cores used, resulting in:

$$O(Q/k)$$

for practical implementations.

d) *Space Complexity:*

The prototype stores both classical and quantum blocks along with metadata. The total space requirement is:

$$O(HW + M)$$

where $O(HW)$ corresponds to the image data and $O(M)$ to block metadata, which is negligible compared to the image size.

Process	Complexity	Remarks
Entropy computation	$O(HW)$	Linear in image size
Block selection	$O(M)$	Entropy-based sorting
SDC simulation	$O(Q)$	Q : quantum blocks
Metadata handling	$O(M)$	Negligible overhead

TABLE I: Space Complexity Summary

e) *Interpretation:*

The overall computational complexity of the system is $O(HW + M \log M)$, dominated by image size and entropy evaluation, while space usage scales linearly with the input.

The prototype uses circuit-level SDC simulation to approximate performance, while the full realization of such image-scale quantum transmission remains a longer-term goal as qubit counts and entanglement distribution technologies continue to advance [14], [15].

V Implementation

V-A Encoding and Transmission:

In the prototype, high-entropy blocks are changed into binary streams. Superdense coding (SDC) maps two classical bits to one qubit, so the streams are padded for even length. The encoding process is done in Qiskit. Alice and Bob share entangled Bell pairs, and Pauli gates are used to encode 2-bit symbols into qubits. Transmission is simulated with noise models [16].

While the framework assumes that the remaining blocks and metadata would be sent over a classical channel, this was not included in the prototype since classical transfer is already well established.

Algorithm 1:

Input: Input image I , block size B , top percentage p , noise parameters

Output: Received quantum bit pairs

1. Load image I and convert to PNG format;
2. Compute entropy score for each block;
3. Select top- $p\%$ high-entropy blocks $\rightarrow$ quantum set;
4. Mask remaining blocks to form classical image;
5. Build quantum noise model and backend simulator (Qiskit Aer);
6. Prepare SDC circuits for encoding bit pairs;

foreach selected block (i, j) in quantum set **do**

- a. Convert block to bitstream, pad to even length, and chunk into 2-bit pairs;
- b. Store bit pairs, block indices, and counts for transmission;

7. Construct metadata M ;
 8. Group bit pairs into batches for parallel processing;
 9. Simulate SDC transmission of all bit pairs with noise model and record received pairs;
 - return** received quantum bit pairs
-

V-B Metadata Handling

Algorithm 2: Metadata Generation and Handling

Input: blocks ($H \times W$ grid of image blocks), top_blocks (list of high-entropy block triples $\langle \text{score}, i, j \rangle$), block_size, top_percentage

Output: Metadata structure M

$M.blocks \leftarrow []$;

$M.info \leftarrow \{block_size : k, top_percentage : p\}$;

foreach block (i, j) in B **do**

if $(i, j) \in S$ **then**

 Append $\{row : i, col : j, type : "quantum"\}$ to $M.blocks$;

else

 Append $\{row : i, col : j, type : "classical"\}$ to $M.blocks$;

return M ;

V-C Receiver-Side Decoding

At the receiver, metadata is read first to help with reconstruction. The quantum payload is decoded from qubits back into bit strings, padding is removed, and compressed quantum blocks are recovered. Classical blocks, presumed transmitted, would be directly decompressed. Both sets of blocks return to their original coordinates to reconstruct the image.

Algorithm 3: Receiver-Side Decoding and Reconstruction

Input: Received quantum bit pairs, metadata M , classical image, block size B

Output: Reconstructed image $\hat{I}$

1. Reconstruct quantum blocks from received bit pairs using indices and counts;
 2. Assemble quantum image from reconstructed blocks with original shape;
 3. Initialize $\hat{I}_{quantum} \leftarrow$ quantum image copy;
 4. **foreach** block (i, j) in top-entropy set **do**
 - a. Extract block region from quantum image;
 - b. Apply non-local means denoising to the block;
 - c. Replace block in $\hat{I}_{quantum}$ with denoised version;
 5. Combine classical and quantum blocks using metadata M to form $\hat{I}$;
 6. Plot histogram of pixel-wise errors and generate heatmap;
- return** $\hat{I}$
-

VI Results and Outcomes

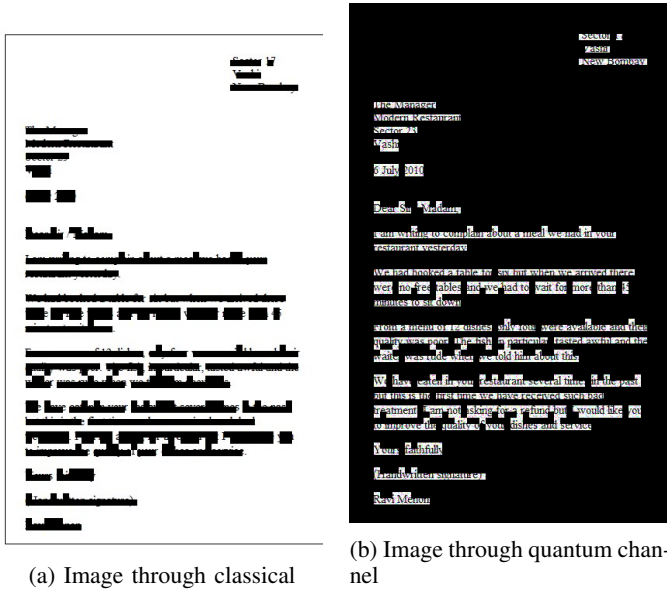
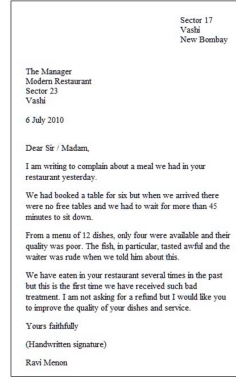


Fig. 1

Figure 1a is for the all-classical design in which every block goes through the classical channel. Similarly, Fig 1b is for the quantum assisted design, where only the blocks with the highest entropy encoding go through SDC. As observed in Fig. 1b, the blocks assigned to the quantum channel cover regions of high information, such as edges and textured areas, confirming the effectiveness of entropy-based block selection.

Reconstructed Image



Error Heatmap (Original vs Hybrid)

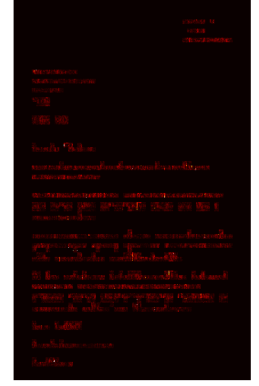


Fig. 2: Comparison of the reconstructed hybrid image and corresponding error heatmap at noise 30%.

In Fig 2, The reconstructed image maintains high structural similarity with minimal localized deviations. The corresponding error heatmap indicates that non-zero differences are primarily concentrated around fine-textured regions and high-contrast edges, which is consistent with the entropy-based block selection strategy. The red colors indicates the difference between original image and reconstructed image.

Resolution	Block Size	PSNR (dB)	SSIM
128 × 128	16 × 16	55.59	0.997
256 × 256	16 × 16	48.22	0.999
512 × 512	16 × 16	48.84	0.996

TABLE II: Image Reconstruction Quality for Hybrid Transmission

Table II represents the reconstruction quality achieved by the proposed hybrid framework across different image resolutions with noise 30%. SSIM remains high because only a small fraction of high-entropy blocks are transmitted via SDC.

Histogram of Pixel Values

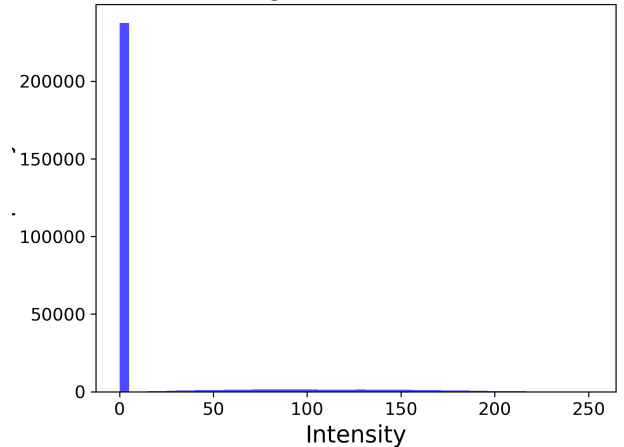


Fig. 3: The histogram of pixel-wise absolute differences between the original and reconstructed image.

In Fig 3, The majority of pixel errors are located around zero, suggesting that the hybrid transmission results in the retention of the majority of visual information with negligible degradation. The only high-error pixels are in detailed and high-entropy sections that simulated quantum noise.

VII Innovation and Novelty

The proposed system introduces a hybrid image transmission framework that combines classical and quantum channels in a block-wise manner. Unlike earlier efforts that focus only on classical compression or quantum communication protocols, our method uses entropy-based block selection. This approach assigns high-information content to the quantum channel and low-information content to the classical channel. It ensures an efficient use of limited quantum resources while keeping reliability through classical transfer.

This selective transmission strategy enables efficient utilization of limited quantum resources while maintaining the reliability and scalability of classical communication. In contrast to SDC-only approaches, the proposed framework avoids unnecessary quantum transmission of low-entropy regions, reducing entanglement consumption and circuit overhead. At the same time, unlike purely classical methods.

The primary novelty of this work lies in its communication-centric hybrid design, which prioritizes practical deployability under current quantum hardware constraints. By combining entropy-driven block selection with classical-to-quantum bit mapping

VIII Use Case Applications

The proposed hybrid framework is clearly capable of addressing real-world problems where both efficiency and reliability are extremely important:

- In bandwidth-limited environments such as deep-space or underwater communication, the hybrid transmission strategy can improve effective information throughput by selectively applying Superdense Coding to high-information image regions without increasing overall bandwidth requirements.
- The framework supports tamper-aware image distribution by allowing authentication data, watermarks, or other critical metadata to be transmitted via the quantum channel, while the bulk image content is delivered classically. This separation enhances data integrity without imposing full quantum encoding overhead.

This hybrid allocation builds a scaling advantage and flexibility into any resource allocation strategy, ensuring quantum resources are used where they provide the most value added and classical channels provide extra stability and throughput.

IX Limitations and Future Work

Entropy limitation, While entropy effectively identifies information-rich regions in images with structured content and smooth backgrounds, it becomes less discriminative for images dominated by dense textures or uniformly complex patterns, such as crowd scenes. In such cases, a large fraction

of blocks may exhibit similarly high entropy values, reducing the effectiveness of selective quantum allocation and limiting potential resource savings.

Quantum hardware scalability and entanglement management constitute a fundamental limitation of the proposed framework. Current quantum devices support only a limited number of qubits with short coherence times and non-negligible gate noise, which restricts the practical use of Superdense Coding for high-resolution image transmission. In addition, the reliable generation, distribution, and synchronization of entangled Bell pairs over long distances remain challenging and are largely confined to laboratory-scale or resource-intensive experimental environments. These combined constraints limit the number of image blocks that can be transmitted quantumly and hinder the scalability of hybrid image transmission systems under present-day hardware conditions.

System-level integration between classical and quantum communication channels is also underexplored. The current prototype is simulation-based and relies on Qiskit noise models rather than execution on physical quantum hardware or quantum networks. As a result, performance evaluation is limited to idealized or approximate conditions and may not fully reflect real-world network behavior. Related work on teleporting digital images demonstrates feasibility only for small-scale or toy examples, further highlighting the gap between simulation and deployment [13].

Future research can address these limitations through several directions. Scalable entanglement distribution techniques, such as quantum repeaters and entanglement purification, are essential for extending transmission distance and reliability. Dynamic block selection policies that adapt to channel conditions, noise levels, and available quantum resources could further improve efficiency. Experimental validation on quantum network testbeds would enable more realistic performance assessment beyond simulation. Additionally, integrating hybrid quantum-classical error correction schemes, including learning-based denoising methods, could enhance robustness against noise and decoherence.

X Conclusion

This research described a hybrid approach to image transmission that combines classical channels and quantum superdense coding (SDC) for efficient and reliable transmission of visual information. Based on decomposing images into entropy-defined blocks, the system encodes only the region of maximum information density with quantum transmission using the SDC protocols, while the remainder of the image with low-entropy and the metadata are transmitted through classical channels. This results in a bandwidth-efficient design that retains essential aspects of the image, while leveraging the different strengths offered by classical and quantum communication.

An initial prototype implementation with Qiskit simulations demonstrated block-wise entropic selection and SDC encod-

ing, followed by metadata-driven reconstruction. The hybrid transmission model achieved 16% bandwidth reduction by sending 20% of high-entropy blocks through the quantum (SDC) channel. The experimental results show nearly lossless reconstruction with a PSNR of 39.3 dB, SSIM of 0.993, and Image fidelity of 99.34%. The hybrid approach increased throughput efficiency from 5 Mbps to approximately 5.8 Mbps (estimated) using the relation. While the time delay for quantum transmission remains limited by simulation time and existing QRNG devices, one take-away for future imaging applications, is that SDC offers the potential for secure and scalable transmission over time.

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